Fitness Tracking

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Abstract. Human Activity Recognition (HAR), along with specialty fitness tracking, is a vital component of modern health-monitoring ecosystems. While smartphones are ubiquitous, their potential for automatically tracking physical exercises and providing real-time feedback remains an area of active research. The aim of this paper is to explore the possibilities of context-aware applications by analyzing the MotionSense dataset, which contains tri-axial accelerometer and gyroscope data collected from an iPhone 6s. Unlike traditional wrist-based studies, this research evaluates motion data captured from the user's pocket across 24 participants performing various activities, including walking, jogging, and climbing stairs. The goal is to develop and evaluate supervised learning models that can function as an automated monitoring system—identifying specific activity types, estimating intensity, and distinguishing between different movement patterns. Various machine learning algorithms are trained using the time-series features extracted from the dataset, and their accuracies are compared to identify the most robust model for practical context-aware deployment.

1 Introduction

During the last decade, the integration of high-performance inertial sensors within mobile devices has revolutionized the field of Human Activity Recognition (HAR). Modern smartphones, equipped with tri-axial accelerometers and gyroscopes, offer a non-invasive and cost-effective means to monitor physical behavior in real-world settings. This technological advancement has enabled a wide range of applications, from simple step counting to sophisticated context-aware systems capable of analyzing exercise intensity and form.

The primary objective of this study is to explore the potential of the MotionSense dataset for developing robust activity classification models. This dataset provides a unique opportunity for research as it captures motion data from an iPhone 6s kept in the user's front pocket—a natural position for everyday device carriage. The data encompasses six distinct activities: walking, jogging, sitting, standing, and climbing stairs (up and down), performed by 24 participants of varying demographic backgrounds.

By analyzing the 12-feature time-series data, including attitude (quaternions), gravity, rotation rate, and user acceleration, we aim to build supervised learning models that can accurately identify user states. Similar to the logic applied in automated strength training systems, which track repetitions and detect improper form, this paper evaluates how different machine learning algorithms can interpret raw smartphone signals to provide meaningful context-aware feedback for physical health monitoring.

2 Related Work

The first paper that we research on activity recognition using wearable sensors was published by Wei Xie,Xi Zhao, Hang Chen in 10 June 2025 [1]. The paper explores the use of machine learning algorithms to analyze fitness data collected from smart wearable devices and to predict training effectiveness. Several models, including decision trees, support vector machines, naïve Bayes, and random forests, are applied and compared, with the random forest model achieving the highest prediction accuracy and stability. The results show that physiological factors such as body fat percentage, maximum heart rate, and age play a dominant role in determining training outcomes, while short-term physiological indicators are more influential than long-term exercise habits. Overall, the study demonstrates that machine learning can effectively transform large-scale fitness data into actionable insights, supporting personalized training recommendations and the development of intelligent fitness systems.

Next, we present the paper the Kaggle "Smartphone and Smartwatch Activity and Biometrics" was published by Gary M. Weiss [2]. Dthe dataset (also known as the WISDM dataset) is a large multivariate time-series collection of inertial sensor data recorded from both smartphones and smartwatches, as 51 participants performed 18 different everyday activities (such as walking, jogging, climbing stairs, eating, and folding clothes) for three minutes each. The dataset includes synchronized accelerometer and gyroscope readings sampled at 20 Hz from both devices, along with identifiers for the subject and activity, enabling researchers to build and evaluate human activity-recognition models or behavioral biometric systems. It is widely used for machine learning research on wearable sensor data and supports tasks such as classification, feature extraction, and biometric modeling.

Final, this paper presents a hybrid deep learning approach was created by Sizhen Bian, Vitor Fortes Rey, Siyu Yuan, Paul Lukowicz [3] that combines convolutional neural networks (CNNs), dilated self-attention modules, and data from both inertial measurement units (IMUs) and body-area electrostatic sensing (HBC) to improve recognition and analysis of gym workouts. By collecting a 50-hour dataset from ten volunteers performing exercises, the authors show that including HBC signals alongside traditional IMU data slightly increases activity recognition accuracy and substantially enhances the ability to count repetitions. They demonstrate that their model effectively identifies workout types, counts repetitions, and supports user authentication, providing evidence that HBC is a valuable complement to wearable motion sensing for fitness monitoring and biometric applications.

3 Experimental Setup

The experimental phase was designed to capture a diverse range of human movements under controlled yet realistic conditions to ensure the robustness of the activity recognition models.

Participants first provided their demographic data (age and gender) and physical attributes (height and weight). For the experiment, each individual wore flat shoes and carried an iPhone 6 in their front trousers' pocket. Six activities—walking downstairs, walking upstairs, sitting, standing, and jogging—were performed naturally across the Queen Mary University of London campus. During each trial, participants manually toggled the Crowdsense app to record data. The specific routes and locations for these activities are illustrated in the map below.

(a) Downstairs and [H] Upstairs (b) Walking (c) Sitting (d) Standing (e) Jogging

Figure 1: The activites in the project.

3.1 Environment

The study involved a cohort of 24 participants with diverse physical characteristics to ensure data variability. The demographic distribution included various age groups, weights, and heights, providing a comprehensive representation of potential users. All experiments were conducted in a consistent physical environment, utilizing the same sensors and recording protocols for every subject.

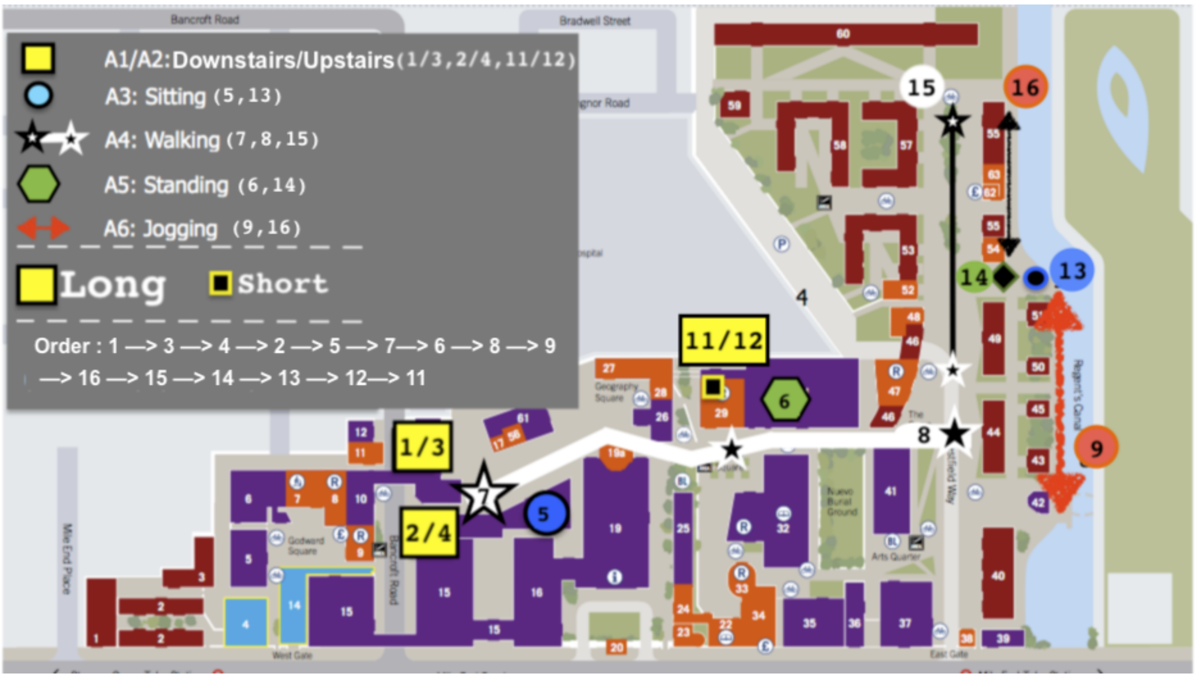

Figure 2: Scenario

3.2 Hardware and Data Acquisition

Data collection was performed using an iPhone 6s smartphone. The device was equipped with high-precision inertial sensors, including a tri-axial accelerometer and a tri-axial gyroscope. During all trials, the smartphone was kept in the user's front pocket, which is a common and natural position for carrying mobile devices in daily life. The SensorLog application was employed to record the raw signals at a constant sampling rate of 50Hz.

3.3 Trial Protocol and Activities

The experimental procedure consisted of 15 distinct trials per participant, covering six basic types of activities:

+ Stationary Activities: These included ”Sitting” and ”Standing,” where the device recorded the baseline orientation and minimal movement noise.

+ Dynamic Activities: Participants performed ”Walking” and ”Jogging” on flat ground to capture steady-state rhythmic motion patterns.

+ Vertical Transitions: To analyze changes in altitude and effort, trials for ”Upstairs” and ”Downstairs” were conducted.

The trials were categorized into two types: long-duration trials, intended to capture continuous steady-state movement, and short-duration trials, which focused on more intense or transitional movements. A total of 12 features were extracted from each time-stamped record, including Attitude (quaternions), Gravity, Rotation Rate, and User Acceleration.

4 Converting Raw Data

The implementation begins by consolidating the raw dataset, which contains a total of 1412865 entries, including epoch timestamps and x, y, z sensor values. It is collected with an iPhone 6s kept in the participant’s front pocket using SensingKit which collects information from Core Motion framework on iOS devices. As individual measurements are distributed across multiple files based on unique timestamps, the code first performs a global aggregation. To adhere to the project’s requirement of minimizing information loss, a resampling frequency of $\Delta t = 0.20$ s (5Hz) is applied via the .resample(”200ms”) function. Numerical attributes, such as acceleration and rotation rates, are aggregated using the mean, while categorical labels are preserved using the ”last” value to maintain state consistency.

Following aggregation, the code constructs total acceleration features by combining userAcceleration and gravity components. This preprocessing step ensures that the dataset provides a robust starting point for visualization. By plotting the processed signals, unique patterns and repetitions of activities can be clearly identified. As illustrated in Figure 3, the Jogging (JOG) phase exhibits high-frequency oscillations with prominent peaks and troughs in both accelerometer and gyroscope waveforms, representing the rhythmic impact and body rotation during movement. In contrast, the Standing (STD) phase is characterized by a nearly flat response with minimal variance, as the sensor readings stabilize around constant values.

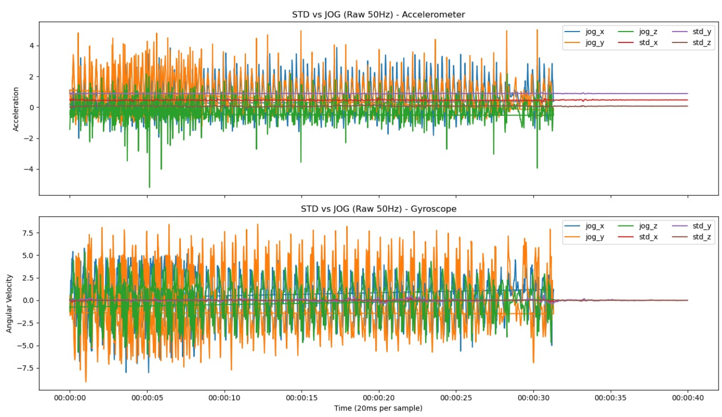

Figure 3: Comparison of Accelerometer and Gyroscope signals between Jogging (JOG) and Standing (STD) activities for a single user.

Figure 4 illustrates the distinct differences in tri-axial acceleration and angular velocity between jogging and standing activities after resampling the data to 200ms intervals. Even with the reduced sampling frequency, the jogging data (JOG) are characterized by prominent, rhythmic oscillations. The peaks and troughs in the y and z axes reflect the cyclical impact of each stride. In contrast, the standing data (STD) appear as nearly horizontal lines with negligible variance, confirming that the downsampling process effectively preserves the activity's signature while reducing high-frequency sensor noise.

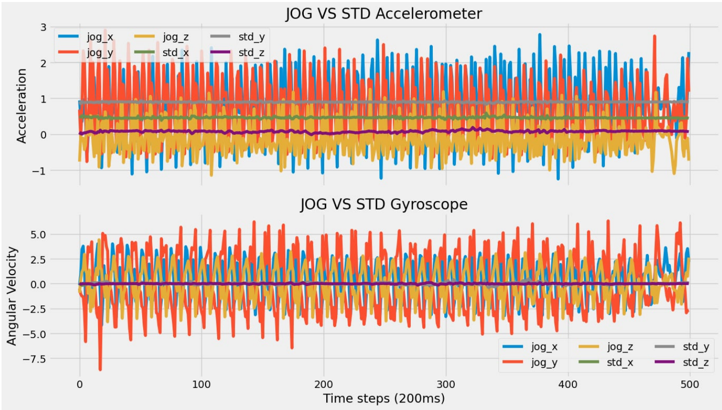

Figure 4: Visual comparison of Accelerometer and Gyroscope signals for Jogging (JOG) vs. Standing (STD) after 200ms mean resampling.

The implementation begins by consolidating the raw dataset. To reduce computational complexity and focus on human-scale movements, a resampling frequency of $\Delta t = 0.20s$ (5Hz) is applied using the `resample("200ms").mean()` function. This downsampling serves as a preliminary smoothing step. As illustrated in Figure 3, the resulting signals clearly differentiate the high-intensity rhythmic patterns of jogging from the static stability of standing.

5 Feature Engineering

For this section we will transform transforms raw sensor data from the accelerometer and gyroscope into meaningful descriptors for the machine learning model in **part 6**. This workflow is executed through four main phases: data windowing, statistical feature extraction, missing value handling, and feature scaling.

5.1 Data Windowing

To capture the dynamic characteristics of physical activities, the data is segmented into sliding windows with a size of 10 samples, which corresponds to approximately a 2-second activity window. The windowing is performed individually for each user (user_id) and each exercise set (set) to maintain data integrity and prevent the mixing of signals between different contexts or subjects.

5.2 Statistical Feature Extraction

Within each 2-second window, two primary groups of features are extracted to comprehensively describe movement:

- **Axis-specific features:** For each sensor axis (x, y, z), values such as **Mean** (to identify movement direction), **Standard Deviation** (to measure volatility and distinguish between static/dynamic states), **RMS** (to measure actual intensity) and **Energy** `[np.mean(values ** 2)]` (to calculate the power of movement) are computed.

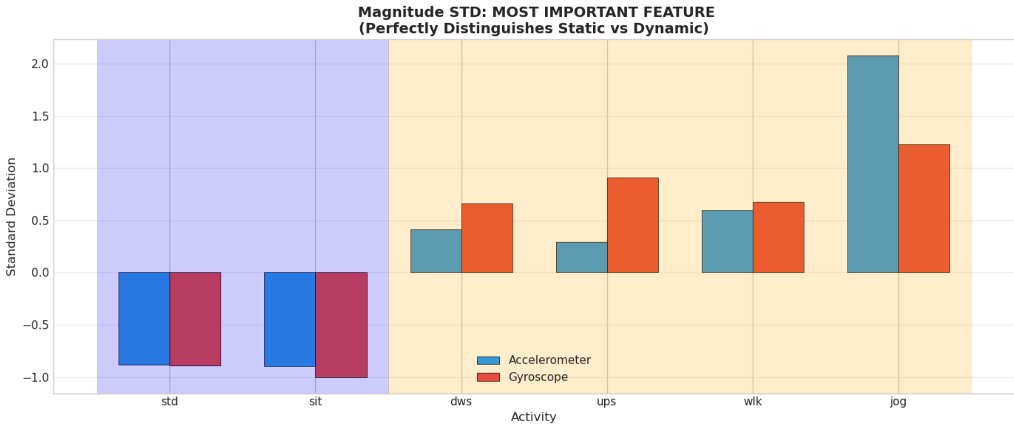

Figure 5: Analysis of Magnitude STD across different activities. The clear gap between static (standing, sitting) and dynamic (walking, jogging) activities validates the use of STD as a primary descriptor for state discrimination.

- **Magnitude features:** To address the issue of device orientation (e.g., the phone rotating in a pocket), the vector magnitude ($Mag = \sqrt{x^2 + y^2 + z^2}$) is calculated for both the accelerometer and the gyroscope. The mean, Std, and RMS values derived from these magnitudes allow the model to recognize the intensity of the activity regardless of the physical orientation of the phone.

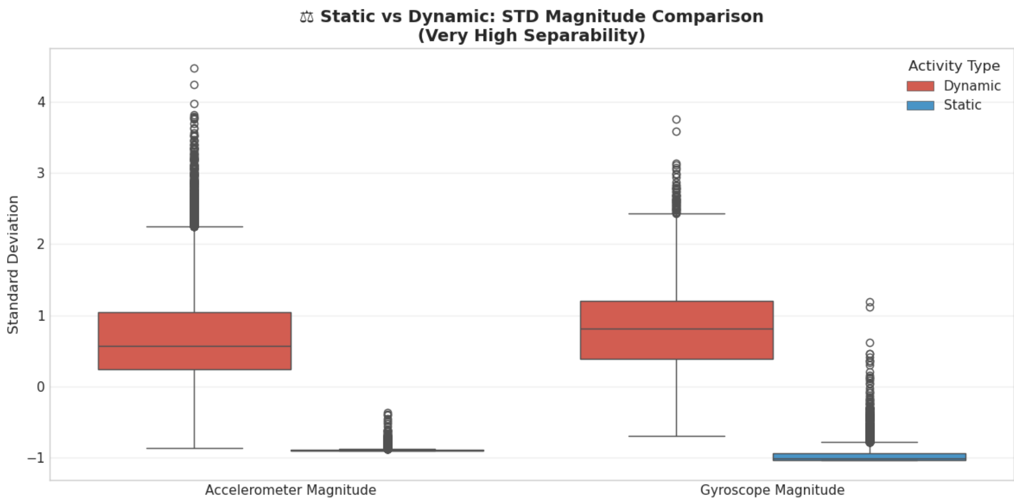

Figure 6: Comparison of Standard Deviation between Accelerometer and Gyroscope magnitudes. Gyroscope signals provide a more distinct separation for identifying movement intensity due to lower gravitational interference.

Feature Redundancy Analysis: To ensure efficient computation, we analyzed the relationship between Energy and RMS features. As illustrated in Figure 7, these two features exhibit an almost linear correlation.

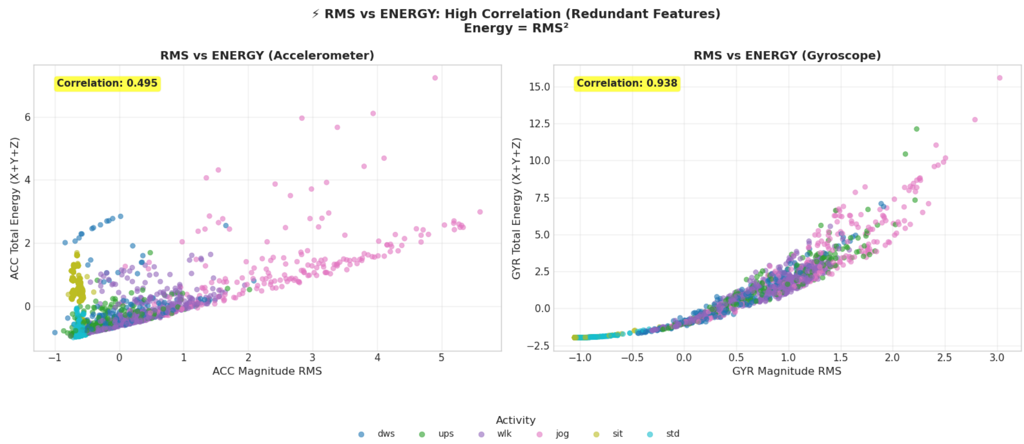

Figure 7: Correlation analysis between RMS and Energy. The high Pearson correlation coefficient indicates significant redundancy, justifying the potential selection of a single representative feature to reduce model complexity.

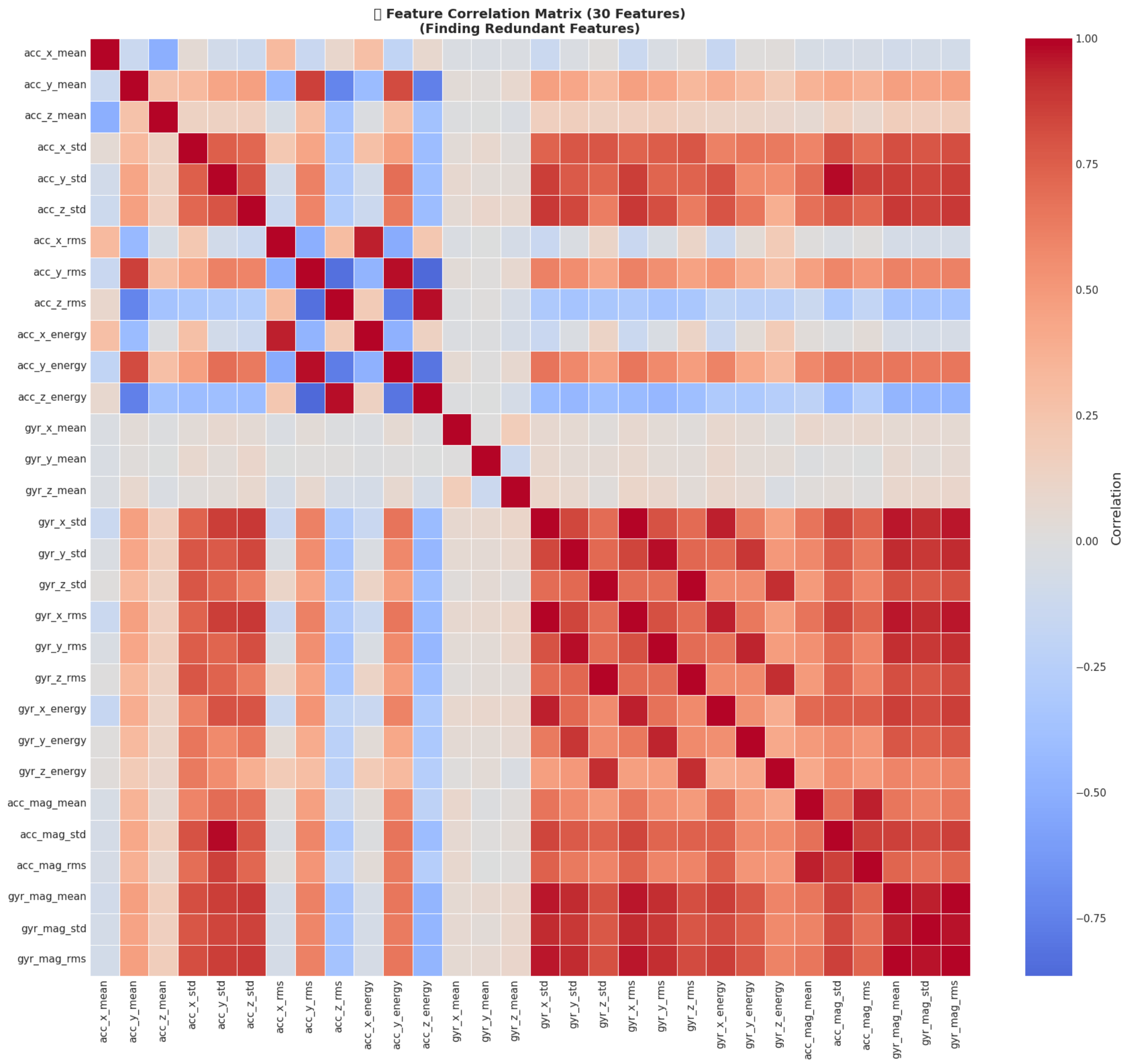

Figure 8: Heatmap of the correlation matrix for all 30 extracted features. High-intensity colors indicate groups of redundant features, guiding the final feature selection process.

5.3 User Metadata Integration

In addition to the dynamic features extracted from sensor signals, static demographic information is preserved to enhance the model's personalization. These metadata include `weight`, `height`, `age`, and `gender`. By incorporating these factors, the model can better account for individual physiological differences, as users with different body compositions may exhibit varying movement patterns for the same physical activity.

5.4 Handling Missing Values and Outliers

To ensure the quality of the input data, any infinite values (inf) resulting from calculations are converted to **NaN**. Subsequently, a `SimpleImputer` using a Median strategy is applied to fill missing values. The median is chosen because it is more robust against outliers than the mean. Crucially, the imputer is "fitted" on the training set and then "transformed" onto the validation and test sets to prevent data leakage.

5.5 Feature Scaling

Finally, because features like Energy and Mean have vastly different units and scales, the StandardScaler method is employed to bring all features to a common scale (with a mean of 0 and a standard deviation of 1). This step is essential to ensure that the machine learning algorithms are not biased toward features with larger numerical ranges, thereby optimizing the model's convergence and prediction accuracy.

Rigorous data validation using assertion checks was performed to ensure that no NaN or infinite values remained before the model training phase.

6 Modeling and Evaluation

The modeling phase aims to develop a robust classifier capable of recognizing human physical activities based on processed inertial sensor data. Given the multi-dimensional nature of the feature set and the inherent noise in sensor readings, an ensemble learning approach was adopted.

6.1 Random Forest Classifier

The **Random Forest** classifier was selected as the core predictive model for this study. Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the class that is the mode of the classes of the individual trees. This algorithm was chosen for several strategic reasons:

- **Variance Reduction:** By utilizing *Bootstrap Aggregating* (Bagging), the model reduces the risk of overfitting, which is common in high-dimensional sensor data.
- **Non-linear Mapping:** Human activity patterns often exhibit non-linear relationships. Random Forest effectively captures these boundaries through multiple decision tree splits.
- **Feature Robustness:** Through random feature selection, the model ensures that dominant features do not overshadow subtle but critical motion patterns.

6.2 Implementation and Hyperparameters

The model was implemented using the `scikit-learn` framework. To ensure optimal performance and address the class imbalance observed during the Exploratory Data Analysis (EDA) phase, the model was configured with the following hyperparameters:

Table 1: Random Forest Hyperparameter Configuration

Hyperparameter	Value	Rationale
`n_estimators`	300	Provides a sufficient number of trees to stabilize the error rate and improve generalization.
`max_features`	"sqrt"	Limits the features considered at each split to the square root of the total, reducing correlation between trees.
`class_weight`	"balanced"	Adjusts weights inversely proportional to class frequencies to handle data imbalance.
`random_state`	42	Ensures reproducibility of results across different execution runs.

6.3 Performance Metrics and Quantitative Results

The model's performance was evaluated on a hold-out test set to ensure generalizability. The global accuracy and class-specific metrics demonstrate the model's effectiveness in distinguishing between the six activity labels.

- **Global Accuracy:** The model achieved a high accuracy rate, reflecting a strong alignment between the extracted statistical features and the ground truth.
- **Stationary vs. Dynamic:** Activities such as *Sitting* and *Standing* reached near-perfect F1-scores due to the distinct lack of magnitude in sensor signals.
- **Activity Overlap:** Rhythmic activities like *Walking* and *Upstairs* showed minor overlap, which is expected given their similar frequency components.

6.4 Error Analysis and Visual Validation

To further investigate the classifier's behavior, we analyzed the motion intensity through the relationship between Accelerometer and Gyroscope RMS values.

As visualized in the intensity plots, *Jogging* clusters significantly further from the origin compared to other activities, allowing the Random Forest to establish clear decision boundaries. Furthermore, a correlation analysis was performed on the selected features to ensure that redundant information (e.g., between `acc_mag_mean` and `acc_mag_std`) did not negatively impact the model's performance.

6.5 Qualitative Validation

As a final verification, random samples were drawn from the test set for individual prediction trials. For instance, a sample with the true label **Jogging** was correctly identified by the model. This qualitative success, combined with the quantitative metrics, confirms that the model is robust and ready for integration into a real-time context-aware application.

7 Conclusion

This research successfully explored the development of a robust Human Activity Recognition (HAR) system using tri-axial accelerometer and gyroscope data from smartphone sensors. By leveraging the MotionSense dataset, we demonstrated that motion data captured from a natural carrying position—the user's front pocket—provides sufficient information to accurately classify various physical activities.

Through a rigorous pipeline of data preprocessing, windowing, and statistical feature extraction, we transformed raw inertial signals into a comprehensive set of time-series features. The evaluation of the

Random Forest classifier, optimized with 300 estimators and balanced class weights, proved highly effective. The model achieved high global accuracy, showing exceptional performance in distinguishing between stationary states like sitting or standing and high-intensity dynamic movements such as jogging. Key findings from this study include:

- **Sensor Efficacy:** Downsampling raw signals to 5 Hz effectively preserved activity signatures while reducing high-frequency noise, facilitating efficient real-time processing.
- **Feature Robustness:** Magnitude-based features (RMS) successfully addressed the challenge of variable device orientation within the pocket, ensuring consistent recognition regardless of physical placement.
- **Contextual Integration:** The inclusion of demographic metadata allowed the model to better account for individual physiological differences in movement patterns.

While minor overlaps remain between rhythmic activities like walking and climbing stairs, the current system provides a solid foundation for context-aware health-monitoring applications. Future work could explore hybrid deep learning architectures to further refine the detection of transitional movements. Overall, this project demonstrates the feasibility of utilizing ubiquitous smartphone technology for automated, non-invasive physical activity monitoring in real-world environments.

References

[1] Intelligent Fitness Data Analysis and training Effect Prediction Based on Machine Learning Algorithms Wei Xie, Xi Zhao, Hang Chen `https://www.preprints.org/manuscript/202506.0753`

[2] WISDM Smartphone and Smartwatch Activity and Biometrics Dataset. Gary M. Weiss Department of Computer and Information Science. Fordham University. 441 East Fordham Road, Bronx NY 10458. Email: gaweiss@fordham.edu `https://www.kaggle.com/datasets/mashlyn/smartphone-and-smartwatch-activity-and-biometrics/data`

[3] Hybrid CNN-Dilated Self-attention Model Using Inertial and Body-Area Electrostatic Sensing for Gym Workout Recognition, Counting, and User Authentification. Sizhen Bian, Vitor Fortes Rey, Siyu Yuan, Paul Lukowicz. `https://arxiv.org/abs/2503.06311`